EviDrug: Supplementary Material

Clinical Evidence-Guided Residual Learning for Medication Recommendation with Dynamic DDI Regularization

This supplement provides the complete cohort definitions, additional model details, experimental settings and seed-level results for the three datasets in the main paper. Its final section lists all 520 SICdb medication-dictionary decisions.

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S1 Data and Cohort Construction

We use MIMIC-IV 3.1 [1], NWICU 0.1.0 [2] and SICdb 1.0.8 [3]. The three cohorts use the complete eligible patient populations. This section specifies their source fields, medication mappings, timing rules, patient partitions and training reference data. All reported counts are cohort-level summaries.

S1.1 Source fields and task units

Table S1: Source fields and prediction units.

Dataset	Source tables and required fields	Prediction unit
MIMIC-IV 3.1	admissions: subject_id, hadm_id, admittime; diagnoses_icd and procedures_icd: hadm_id, icd_code, icd_version, seq_num; prescriptions: hadm_id, ndc	Final eligible hospital admission; all earlier eligible admissions provide history
NWICU 0.1.0	admissions: subject_id, hadm_id, admittime, dischtime; diagnoses_icd: icd_code, icd_version, seq_num; prescriptions: hadm_id, ndc, starttime, stoptime	Final eligible hospital admission after adjacent non-overlap filtering; hospital-level cohort
SICdb 1.0.8	cases: CaseID, PatientID, ICD10Main, OffsetAfterFirstAdmission, TimeOfStay; d_references: ReferenceName, ReferenceGlobalID, ReferenceValue; medication: CaseID, DrugID, Amount, AmountPerMinute, Offset	Final eligible ICU/surgery case; earlier cases must finish before that query starts

Current inputs are whole-encounter clinical codes, together with strictly earlier medication sets. Labels are sets of ATC3 classes over the full encounter. The task is encounter-level medication-set reconstruction. Dose, route, drug administration order and adverse events are not prediction targets. Medication names in the frozen mapping dictionary describe public drug semantics and contain no patient frequencies.

Diagnosis and procedure tokens retain their namespaces and ICD version. MIMIC-IV has both diagnoses and procedures. NWICU has diagnosis codes; its released ICU procedure-event table is not a hospital procedure-ICD table and is not used as a surrogate. SICdb uses the principal ICD10Main diagnosis only. Missing procedure modalities have no real procedure codes; the baseline-compatible UNK placeholder does not contribute association evidence.

S1.2 Eligible cohort statistics

Table S2: Complete eligible cohorts and training vocabularies.

Cohort	Patients	Encounters	Mean encounters	Mean final target ATC3	Train code entries	Train ATC3
MIMIC-IV	86,670	352,967	4.0725	21.8227	35391	201
NWICU	12,492	43,115	3.4514	11.9402	6258	167
SICdb	3,898	9,550	2.4500	9.7863	1069	81

Code entries include PAD, UNK and GAP. The code vocabulary and medication output vocabulary are fitted to training patients only. The all-cohort medication universes are 201, 170 and 83 for MIMIC-IV, NWICU and SICdb, whereas the respective model output sizes are 201, 167 and 81. These denominators serve different purposes.

MIMIC-IV has a maximum of 98 retained encounters per patient. Its mean medical-code count is 61.5525 per patient across all retained encounters, not per encounter. NWICU encounter counts have median 3, interquartile range 2–4 and maximum 37. Its final target has median 11 ATC3 classes, interquartile range 8–15 and range 1–52; final diagnosis-code counts have mean 4.8204, median 4, interquartile range 2–6 and range 1–87. Of NWICU's final query admissions, 4,958 contain an ICU stay; eligibility does not require an ICU stay.

S1.3 MIMIC-IV medication mapping and eligibility

Normalize reference and source NDCs by removing a terminal .0, requiring a nonzero numeric string and padding to 11 digits. The frozen MIMIC rule unions all mapped ATC3 classes for a normalized NDC. ATC3 is the first four characters of the source ATC code. Clinical-code ordering follows source seq_num within diagnosis and procedure namespaces, with diagnoses serialized before procedures.

Begin with patients having at least two source admissions. Retain admissions with diagnosis codes and mapped medication labels; procedures may be empty. Sort by admission time and then admission ID; require at least two retained admissions. The recorded filtering counts are 380 admissions without diagnosis and 58,499 without mapped ATC3 among the repeated-admission preparation pool. These are sequential filters. The original aggregate metadata does not record a full raw-admission denominator for this preparation, so those counts are not presented as a complete source attrition flow. This MIMIC pipeline does not apply prescription/admission overlap checks or discharge-time history filtering.

S1.4 NWICU NDC and temporal audit

The archive has 25,923 patients and 61,843 hospital admissions. Of these, 13,833 patients have at least two admissions, accounting for 49,753 admissions. The diagnosis table has 371,807 ICD-10 rows covering 60,973 admissions and 8,086 distinct versioned codes. Admission IDs are unique; the audit finds zero orphan admission links, patient/admission mismatches or invalid admission intervals in these joins.

Source package NDCs must be an explicit 11-digit nonzero code or a three-segment package code with lengths 4-4-2, 5-3-2, 5-4-1 or 5-4-2. Pad each segment to 5-4-2. Reject incomplete/product-only codes and unsegmented 10-digit strings rather than guessing segment boundaries. Reference NDCs use the numeric normalization above. Accept only a unique ATC3 class per normalized NDC; use no medication-name fallback. The reference table has 672,734 uniquely mapped and 119,121 ambiguous normalized NDCs.

Table S3: NWICU prescription mapping and interval filtering.

NWICU audit quantity	Rows
All source prescription rows	1,852,983

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Table S3 (continued)

NWICU audit quantity	Rows
Nonempty NDC rows	1,632,392
Standardizable package NDC rows	1,591,906
Unique ATC3 mapping before interval checks	1,390,310
Any mapping, including ambiguous	1,517,039
Ambiguous mapping rows	126,729
Order stop earlier than start	136,261
Valid interval outside admission	88,488
Valid interval overlapping admission	1,628,234
Unique mapped and admission-overlapping	1,220,787
Repeated-patient prescription rows	1,410,856
Repeated-patient unique mapped rows	1,063,622
Repeated-patient valid admission-overlapping rows	1,238,661
Repeated-patient mapped admission-overlapping rows	933,613

The 75.03% mapping coverage is 1,390,310 / 1,852,983 source prescription rows, before the interval filter. The final mapped and interval-qualified count is 1,220,787; it has a different denominator. Start and stop must parse, stop must be no earlier than start, and the closed order interval must intersect the admission: start <= discharge and stop >= admission. These labels describe medication order types, not verified administrations.

Retain admissions with diagnosis codes and mapped qualified orders. Exclude a patient if eligible admissions share a start time; the frozen cohort has zero such exclusions. Starting from the latest eligible admission, traverse backward and retain an earlier admission only if its discharge is no later than the next selected admission's start. This removes 20 overlapping history admissions. Require at least two selected admissions. The selected cohort has 12,492 patients and 43,115 admissions, with 7,024 distinct diagnosis codes and 170 medication classes before training-only vocabulary fitting. NWICU's global uniquely mapped source set has 172 ATC3 classes; this is not the cohort or model output vocabulary.

S1.5 SICdb drug dictionary and temporal audit

Only d_references entries with ReferenceName equal to Drug qualify. Medication rows must belong to a repeated-patient case, have positive Amount or AmountPerMinute, and have a finite Offset in [0, TimeOfStay]. Non-drug food/fluid references are excluded before computing drug mapping coverage.

The medication dictionary contains 520 names, 303 with an accepted mapping and 217 unresolved. Mapping uses exact generic names and explicit salt/terminal-e orthographic aliases against the ATC table. A mixture with multiple recognized ingredients must have a unique common ATC3 class. Recorded RxNorm resolutions require a unique normalized ingredient and direct unique ATC membership. Ambiguous classes and edit-distance fuzzy matches are rejected. The dictionary is fixed before model training. Every name-to-class decision appears in the final section.

Table S4: SICdb medication filtering and mapping coverage.

SICdb audit quantity	Count and denominator
All raw medication rows	5,141,127
Non-drug reference rows removed within repeated-patient rows	919,865
Positive-amount/rate, in-case drug rows among repeated patients	863,806
Mapped rows among those qualifying drug rows	697,025
Mapping coverage	697,025 / 863,806 = 80.6923%
Cases missing diagnosis or mapped medication labels	105
Overlapping history cases removed relative to the final query	29

The 919,865 non-drug rows are counted after restricting to repeated patients, whereas 5,141,127 is the full raw medication-file count. They cannot be subtracted directly to infer a complete raw-file attrition flow.

Uppercase ICD10Main and remove whitespace while retaining punctuation; prefix it with D10:. Sort cases by OffsetAfterFirstAdmission and case ID. Start and duration must be finite and duration nonnegative; retain cases with diagnosis and mapped medications. Exclude tied starts. The last eligible case is the query, and every history case must end by that query's start. Unlike NWICU's backward adjacent filter, SICdb histories may overlap each other while each finishes before the query. Require at least one history case.

S1.6 Patient partitions, unknowns and sequence scope

Randomly permute the deterministically ordered patients with partition seed 0. The first floor(0.8N) patients form the development partition; the first floor(0.9 floor(0.8N)) patients train and the remainder validate. Remaining patients test. This gives approximately 72%/8%/20%, with cohort-specific integer rounding. Optimization seeds 0/1/2 do not change patient partitions.

Table S5: Patient partitions, unknown-code occurrences and interaction graphs.

Cohort	Train/val/test patients	Unknown code occurrences, train/val/test	Unseen drug occurrences, train/val/test	Undirected DDI edges
MIMIC-IV	62402/6934/17334	0/1199/2839	0/0/0	631
NWICU	8993/1000/2499	0/343/672	0/5/3	536
SICdb	2806/312/780	0/74/188	0/4/4	206

Code IDs are PAD=0, UNK=1 and GAP=2 followed by sorted training-observed codes. Unknown clinical codes map to UNK. Medications absent from the training output vocabulary are omitted from labels and counted as unseen occurrences; these are occurrence counts, not numbers of affected patients.

There is no patient-count subsampling or preparation-time visit/code truncation. The encoder retains the most recent 512 serialized code positions, independently of cohort eligibility. Visit positions clip at 63 and within-visit positions at 127. Donor encounters share the same structured encoder. Query medications are supervision and evaluation labels, never prediction inputs.

Each validation/test patient has one final-encounter query. Training uses one eligible prefix per patient per epoch, cycling deterministically through that patient's eligible targets. NWICU has 21,873 eligible training prefixes and 8,993 targets per epoch; SICdb has 4,116 and 2,806, respectively. The training monitor uses a fixed first-epoch prefix schedule and no validation/test labels.

S1.7 Fixed training evidence and DDI resource

Training-case donors use training patients only. A training query excludes its own patient's transitions; the additional exclusion under the fixed donor ordering is retained across experiments. The eight-neighbor shortlist interleaves state and clinical-change Jaccard rankings with deterministic tie breaks. Similarity uses clinical codes, never query medications. Clinical association statistics count binary visit-level code/medication co-occurrence, and a training query removes all statistics of its own patient. Validation and test use fixed training references.

The DDI graph uses the public PyHealth DDI_GAMENet.csv resource described by GAMENet as TWOSIDES-derived interaction knowledge. Truncate ATC codes to four characters, retain pairs of distinct training-vocabulary classes, collapse duplicate pairs, symmetrize and zero the diagonal. The three graphs contain 631, 536 and 206 undirected edges. This fixed interaction proxy differs from the training co-prescription adjacency; the selected EviDrug does not use a medication-graph representation branch.

The medication reference tables are NDC_to_ATC.csv and ATC.csv; interaction pairs come from DDI_ GAMENet.csv. All three are the public reference resources distributed with PyHealth. These resources define drug-code mappings and interactions independently of the patient partitions.

Interaction pairs follow GAMENet [4] and TWOSIDES [5], using the reference table distributed with PyHealth [6]. The NDC–ATC and ATC reference tables are available at `https://storage.googleapis.com/pyhealth/resource/NDC_to_ATC.csv` and `https://storage.googleapis.com/pyhealth/resource/ATC.csv`; interaction pairs are at `https://storage.googleapis.com/pyhealth/resource/DDI_GAMENet.csv`.

S2 Additional Method Details

This section completes the definitions of the clinical-memory backbone and medication-specific refinement used in the main paper. All representations are learned independently in each cohort.

S2.1 Structured visit-aware encoding

Each clinical token sums five embeddings: code identity, absolute position, encounter position, within-encounter order and a shared segment embedding. Each embedding has width 192. Layer normalization and dropout 0.1 produce token states. Encounter positions saturate at 63 and within-encounter positions at 127. Diagnosis and procedure namespaces retain distinct code identities.

The most recent 512 serialized code positions are retained. Within each encounter, states are summed and divided by the square root of the retained token count. A $192 \to 128$ projection and the log-transformed code count feed a $129 \to 256 \to 128$ MLP with GELU and dropout 0.35. An empty encounter has a zero state. Query and donor encounters share the encoder.

Each candidate medication has a learned width-32 identity. Its cross-attention reads current-encounter clinical states through separate bias-free $128 \to 32$ key and value projections; values pass through tanh. Attention is normalized over available clinical codes, giving a zero local state for an empty code set. A separate width-32 readout for each medication adjusts its start and continuation logits.

S2.2 Historical context and the patient prior

The history attention in the main paper returns an attention-weighted clinical state $\widetilde{\mathbf{h}}$ and medication-presence profile $\mathbf{f}$. The immediately preceding prescription is $\mathbf{a}$. Context concatenates the current state, preceding state, their difference, the history state and projected historical medications:

$$\mathbf{u} = \left[\mathbf{h}_T; \mathbf{h}_{T-1}; \mathbf{h}_T - \mathbf{h}_{T-1}; \widetilde{\mathbf{h}}; P_m(\mathbf{a}+\mathbf{f})/2\right]. \tag{S1}$$

Each component has width 128. Two scalar features complete the input to the context MLP:

$$\mathbf{c} = f_c([\mathbf{u}; \log(1+T-1); \|\mathbf{a}\|_1/M]). \tag{S2}$$

The MLP has dimensions $642 \to 256 \to 128$, with GELU and dropout 0.35. A $128 \to 256 \to 2M$ MLP produces start and continuation scores, augmented by the medication-specific local readouts. The binary preceding-use indicator a_j selects the appropriate score before applying the sigmoid. Missing historical inputs are zero.

S2.3 Medication-specific case attention

The case bank consists of adjacent encounter transitions from training patients. Current-code and clinical-change Jaccard rankings are interleaved, deduplicated and truncated to eight cases. Clinical-change similarity contributes to shortlisting; current-code similarity additionally contributes to learned attention. Query medication labels do not enter retrieval.

The width-32 attention query includes the current clinical state and medication-local evidence:

$$\mathbf{q}_j^M = (W_q[\mathbf{h}_T; \mathbf{0}; \mathbf{0}] + \mathbf{z}_j) \odot (1 + 0.5\tanh(\mathbf{r}_j)). \tag{S3}$$

Here $\mathbf{r}_j$ is a learned medication-specific metric vector. Donor keys project $[\mathbf{h}_r^+; \mathbf{0}; \mathbf{0}]$ to width 32. Scaled query–key matches plus a learned nonnegative multiple of current-code Jaccard similarity produce the donor-attention weights.

Their values are donor-current medication indicators. Thus attention weights cases separately for each candidate medication.

A nonempty shortlist has total attention mass one; an empty shortlist has mass zero and returns the patient prior. The context-dependent fusion strength is $4\sigma(f_s(\mathbf{c})_j)$. Before taking its logit, the fusion probability is clipped to $[10^{-6}, 1-10^{-6}]$. The memory-supervision derivative in the main paper applies before clipping; a saturated clip has zero derivative.

S2.4 Medication interaction and refinement

The medication-set block has width 128, four attention heads, feed-forward width 256, GELU, dropout 0.1 and pre-normalization. Its $128 \to 128 \to 1$ readout supplies a bounded correction to the memory logit. The block's identity projection is reused in the refinement head.

Association statistics count binary visit-level code–medication co-occurrence, with smoothing $b = 0.1$. For each query-code group G (all codes, diagnoses or procedures), let $s_{jG} = \sum_{c \in G} L_{jc}$. The two features are

$$\phi_{jG}^{\text{std}} = \frac{s_{jG} - \bar{s}_G}{\max(10^{-3}, \text{SD}_j(s_{jG}))}. \tag{S4}$$

$$\phi_{jG}^{\text{avg}} = \frac{s_{jG}}{\max(1, |G|)}. \tag{S5}$$

Standard deviation uses the sample definition. Empty-group features are zero, and special clinical tokens do not enter the counts. The three groups yield six features; procedure features are zero for NWICU and SICdb. Training queries subtract their patient's encounters from all association counts. Validation and test use the complete training counts.

The predictive evidence vector contains the backbone logit, patient prior, memory mass and fusion strength. History-conditioned medication identity scales the shared identity projection by $1 + \gamma(a_j + f_j)/2$, with γ initialized to 0.1. Local, predictive and association projections bring all branches to width 128. A shared vector gate reads these branches together with the frozen medication state. Layer normalization follows gated residual addition.

The final refinement MLP is $128 \to 64 \to 1$ with ReLU. Its final layer starts at zero, and $2\tanh$ bounds the logit correction to $[-2, 2]$. The head has 112,514 trainable parameters. The backbone remains fixed throughout refinement and DDI-feedback optimization.

S2.5 Prediction objective and feedback

The ranking term averages over all valid positive–negative medication pairs in a minibatch. If $\mathscr{P}_B$ contains those pairs, it is

$$\mathscr{L}_{\text{rank}} = \frac{\sum_{(i,j,k) \in \mathscr{P}_B} \text{softplus}(\ell_{ik} - \ell_{ij})}{\max(1, |\mathscr{P}_B|)}. \tag{S6}$$

Medication j is observed and k is unobserved. The loss is zero when there is no valid pair. Prediction uses BCE plus $0.05\mathscr{L}_{\text{rank}}$; backbone learning additionally uses prior BCE with weight 0.25.

Soft selection uses threshold 0.3 and temperature 0.25. The DDI objective pools interacting-pair mass over the minibatch and divides by total unordered-pair mass floored at one. The hard training monitor fixes one sampled prefix per training patient. The target is 90% of its initial interaction rate. The DDI coefficient starts at one, changes once per epoch with step size 20 and is clipped to $[0, 10]$. If the initial monitor rate is zero, the feedback penalty is disabled. Feedback changes training regularization; the inference architecture is unchanged.

S3 Experimental Protocol

S3.1 Optimization and model selection

All model parameters are trained from scratch within each cohort. Optimization seeds are 0, 1 and 2. Each run uses 20 epochs of backbone learning, 20 epochs of evidence refinement and 20 epochs of DDI-feedback optimization.

Batch size is 64 and arithmetic is FP32. AdamW [7] optimizes the embedding encoder at 0.00005 and the remaining backbone at 0.0007. Refinement and feedback optimize the head at 0.0005. Weight decay is 0.001, gradient-norm clipping is 5, and every stage starts a fresh optimizer.

One eligible prefix is sampled per training patient per epoch, cycling through that patient's targets in a deterministic seed-specific order. This gives patients equal sampling weight. Validation and test each use the final eligible encounter. Refinement initialization uses seed s + 4100, feedback optimization uses s + 7100, and the donor ordering seed is 20261026. These optimization choices leave the patient partition unchanged.

Backbone and refinement models maximize validation AP, with the earliest epoch breaking ties. Feedback optimization includes the initial refinement model as epoch 0. Eligible models have AP at most 0.002 below the reference, F1 and Jaccard at most 0.005 below it, and a mean predicted medication count between 0.9 and 1.1 times the reference. Among eligible epochs, selection minimizes validation DDI, then prefers higher AP and an earlier epoch. Feedback uses training queries; model selection uses validation patients.

S3.2 Stage, component and regularization controls

The extended-backbone control gives the backbone 20 additional epochs. Evidence refinement instead fixes the backbone and trains the evidence head. Prediction-only training and all DDI controls start from the same seed-specific refinement model, use fresh optimizers and receive the same 20-epoch budget and selection criteria. Fixed coefficients are 0, 0.05, 0.1 and 1.

History removal uses current-encounter evidence and state-only retrieval. Memory removal eliminates retrieval and donor encoding. Set-block removal eliminates medication-set attention. These three variants learn their own backbone and refinement models. Association removal shares the full backbone and refits the affected head. Each variant is evaluated both before and after DDI-feedback optimization. All variants use the same three optimization seeds.

S3.3 Metrics and analysis groups

Primary AP is average precision over candidate medications for each patient, followed by a patient average; tied scores are grouped. F1 and Jaccard are also patient averages. The common probability threshold is 0.3; COGNet retains its native no-repeat beam decoder with beam width 4 and at most 45 steps. Hard DDI pools interacting unordered predicted pairs over patients and divides by all predicted unordered pairs, with denominator at least one. Interacting pairs per patient is reported separately. DDI measures the fixed ATC3 interaction graph.

Continued candidates have appeared in any preceding eligible encounter; new-to-record candidates have not. Their precision, recall and F1 are micro-averages over patient–medication pairs, and Avg F1 is the arithmetic mean of Cont and New F1. These analysis categories differ from the backbone's immediately preceding-use indicator.

History depth uses 1–2, 3–5 and at least 6 preceding encounters. Prior medication burden uses 0–5, 6–10 and at least 11 classes in the immediately preceding prescription. Group membership is shared across models; groups with fewer than 30 patients are marked descriptive. Association support is the maximum training diagnosis–medication co-occurrence count for a candidate, grouped as 0, 1–2, 3–9 and at least 10. AP within support bins pools patient–medication candidate pairs. Medication-frequency quartiles use training counts and place tied counts in the lower bin.

Threshold sensitivity uses 0.1 through 0.9 with fixed model weights. Equal-size top-k analysis rounds the training prescription-size quantiles 0.25/0.5/0.75/0.9 to positive integers and removes duplicates. Methods share the resulting values within each cohort. Patient-paired bootstrap uses 2,000 resamples and seed 20261103, reusing each resample across models. Intervals average the three EviDrug models against the same fitted SSPNet baseline and describe patient-sampling variability for those fitted models.

S3.4 Baseline budgets and efficiency measurement

The twelve baselines use their recorded single-seed budgets; EviDrug reports three optimization seeds. MetaDrug includes self-adaptation, peer-adaptation and uncertainty filtering, with filtering coefficient 0.2. Its 20 training epochs and 10 auxiliary epochs are preserved. The baseline budgets are listed below.

Table S6: Baseline training budgets, shared across the three cohorts.

Method	Training epochs	Auxiliary epochs	Seed
RETAIN [8]	20	0	0
Deepr [9]	20	0	0
Transformer [10]	20	0	0
GAMENet [4]	40	0	0
MeLU [11]	30	0	0
MICRON [12]	20	0	0
SafeDrug [13]	50	0	0
COGNet [14]	200	0	0
MoleRec [15]	50	0	0
ARCI [16]	20	0	0
SSPNet [17]	200	0	0
MetaDrug [18]	20	10	0

Efficiency uses 128 validation patients per cohort, FP32, one CPU thread, batch size 16, one warm-up and three measured passes. The host is an Intel Xeon Platinum 8352V at 2.10 GHz. The reported time is median pass duration divided by 128. Method order is forward, reversed and rotated by six positions across passes. Timing includes online retrieval, evidence preparation, collation, prediction, native adaptation or decoding, and final selection. Loading and construction of fixed training references precede measurement. All 39 method–cohort measurements and their three pass durations accompany the PDF.

S3.5 Evaluation setting

Models are selected on validation patients before test prediction. The study compares methods within each independently trained cohort. Baselines are existing single-seed runs with their native training budgets, and earlier MIMIC-IV test comparisons informed model development. The reported standard deviations therefore describe EviDrug optimization seeds on the fixed patient partition; they do not estimate baseline training variability.

S4 Results for Individual Optimization Seeds

Tables in this section report test AP, F1, Jaccard and DDI for seeds 0, 1 and 2. Summary rows give the mean and sample standard deviation ($n = 3$, denominator $n - 1$). All seeds use the same patient partition. Each comparison uses the training and selection rules in Section S3. The accompanying tables contain all 144 EviDrug stage/variant/seed results and all 36 baseline results. Shared full-model rows are repeated where needed to make each comparison self-contained.

S4.1 Main model

Table S7: EviDrug main-model test results by optimization seed.

Cohort	Seed	AP	F1	Jaccard	DDI
MIMIC-IV	0	0.7767	0.6428	0.4961	0.0511
	1	0.7761	0.6455	0.4993	0.0518
	2	0.7760	0.6472	0.5011	0.0538
	Mean $\pm$ *SD*	**0.7763** $\pm$ 0.0004	**0.6452** $\pm$ 0.0022	**0.4988** $\pm$ 0.0026	**0.0522** $\pm$ 0.0014

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Table S7 (continued)

Cohort	Seed	AP	F1	Jaccard	DDI
NWICU	0	0.6913	0.6036	0.4524	0.1077
	1	0.6901	0.6007	0.4497	0.1016
	2	0.6912	0.6034	0.4525	0.1035
	Mean ± *SD*	**0.6909** ± 0.0007	**0.6026** ± 0.0016	**0.4515** ± 0.0016	**0.1043** ± 0.0032
SICdb	0	0.7497	0.6414	0.4943	0.1125
	1	0.7517	0.6408	0.4928	0.1194
	2	0.7502	0.6451	0.4979	0.1225
	Mean ± *SD*	**0.7505** ± 0.0010	**0.6425** ± 0.0023	**0.4950** ± 0.0026	**0.1181** ± 0.0052

S4.2 Backbone, refinement and feedback

This comparison separates additional backbone training, evidence refinement with a fixed backbone, and DDI-feedback optimization. Prediction-only training and feedback begin with the same seed-specific refinement model.

Table S8: MIMIC-IV: stage comparison by seed.

Variant	Seed	AP	F1	Jaccard	DDI
Backbone	0	0.7530	0.6320	0.4833	0.0609
	1	0.7534	0.6350	0.4864	0.0624
	2	0.7537	0.6342	0.4854	0.0600
	Mean ± *SD*	0.7534 ± 0.0003	0.6337 ± 0.0015	0.4851 ± 0.0016	0.0611 ± 0.0012
Extended backbone	0	0.7564	0.6384	0.4896	0.0640
	1	0.7561	0.6378	0.4895	0.0628
	2	0.7572	0.6387	0.4909	0.0599
	Mean ± *SD*	0.7566 ± 0.0005	0.6383 ± 0.0005	0.4900 ± 0.0008	0.0622 ± 0.0021
Evidence refinement	0	0.7777	0.6466	0.4997	0.0595
	1	0.7768	0.6469	0.4999	0.0588
	2	0.7776	0.6510	0.5048	0.0602
	Mean ± *SD*	0.7774 ± 0.0005	0.6482 ± 0.0025	0.5015 ± 0.0029	0.0595 ± 0.0007
Prediction-only training	0	0.7775	0.6453	0.4987	0.0574
	1	0.7771	0.6440	0.4973	0.0570
	2	0.7761	0.6482	0.5026	0.0578
	Mean ± *SD*	0.7769 ± 0.0007	0.6459 ± 0.0021	0.4995 ± 0.0027	0.0574 ± 0.0004
EviDrug	0	0.7767	0.6428	0.4961	0.0511
	1	0.7761	0.6455	0.4993	0.0518
	2	0.7760	0.6472	0.5011	0.0538
	Mean ± *SD*	**0.7763** ± 0.0004	**0.6452** ± 0.0022	**0.4988** ± 0.0026	**0.0522** ± 0.0014

Table S9: NWICU: stage comparison by seed.

Variant	Seed	AP	F1	Jaccard	DDI
Backbone	0	0.6722	0.5914	0.4400	0.1186
	1	0.6756	0.5915	0.4405	0.1187
	2	0.6744	0.5910	0.4393	0.1125
	Mean ± *SD*	0.6740 ± 0.0017	0.5913 ± 0.0003	0.4399 ± 0.0006	0.1166 ± 0.0036
Extended backbone	0	0.6752	0.5934	0.4422	0.1187
	1	0.6778	0.5908	0.4394	0.1128
	2	0.6778	0.5953	0.4446	0.1184
	Mean ± *SD*	0.6769 ± 0.0015	0.5932 ± 0.0023	0.4421 ± 0.0026	0.1166 ± 0.0033
Evidence refinement	0	0.6904	0.6042	0.4530	0.1144
	1	0.6904	0.6053	0.4546	0.1163
	2	0.6907	0.6064	0.4558	0.1202
	Mean ± *SD*	0.6905 ± 0.0002	0.6053 ± 0.0011	0.4545 ± 0.0014	0.1170 ± 0.0029
Prediction-only training	0	0.6914	0.6034	0.4518	0.1118
	1	0.6913	0.6014	0.4499	0.1119
	2	0.6921	0.6036	0.4523	0.1144
	Mean ± *SD*	0.6916 ± 0.0004	0.6028 ± 0.0012	0.4513 ± 0.0013	0.1127 ± 0.0015
EviDrug	0	0.6913	0.6036	0.4524	0.1077
	1	0.6901	0.6007	0.4497	0.1016
	2	0.6912	0.6034	0.4525	0.1035
	Mean ± *SD*	**0.6909** ± 0.0007	**0.6026** ± 0.0016	**0.4515** ± 0.0016	**0.1043** ± 0.0032

Table S10: SICdb: stage comparison by seed.

Variant	Seed	AP	F1	Jaccard	DDI
Backbone	0	0.7434	0.6438	0.4966	0.1282
	1	0.7439	0.6459	0.4991	0.1304
	2	0.7433	0.6453	0.4984	0.1299
	Mean ± *SD*	0.7435 ± 0.0003	0.6450 ± 0.0011	0.4980 ± 0.0013	0.1295 ± 0.0012
Extended backbone	0	0.7452	0.6442	0.4972	0.1306
	1	0.7441	0.6420	0.4946	0.1276
	2	0.7462	0.6457	0.4987	0.1307
	Mean ± *SD*	0.7452 ± 0.0011	0.6440 ± 0.0019	0.4968 ± 0.0021	0.1296 ± 0.0017
Evidence refinement	0	0.7506	0.6459	0.4986	0.1254
	1	0.7515	0.6480	0.5014	0.1280
	2	0.7502	0.6495	0.5028	0.1334
	Mean ± *SD*	0.7507 ± 0.0007	0.6478 ± 0.0018	0.5009 ± 0.0021	0.1289 ± 0.0041
Prediction-only training	0	0.7510	0.6446	0.4975	0.1252
	1	0.7515	0.6436	0.4959	0.1251
	2	0.7514	0.6472	0.5001	0.1278
	Mean ± *SD*	0.7513 ± 0.0002	0.6451 ± 0.0019	0.4978 ± 0.0021	0.1260 ± 0.0015

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Table S10 (continued)

Variant	Seed	AP	F1	Jaccard	DDI
EviDrug	0	0.7497	0.6414	0.4943	0.1125
	1	0.7517	0.6408	0.4928	0.1194
	2	0.7502	0.6451	0.4979	0.1225
	Mean ± *SD*	**0.7505** ± 0.0010	**0.6425** ± 0.0023	**0.4950** ± 0.0026	**0.1181** ± 0.0052

S4.3 Major-component ablations

The same removals are evaluated after evidence refinement and after DDI-feedback optimization. This separates the predictive role of each component from its behavior under interaction regularization.

Table S11: MIMIC-IV: component ablations after evidence refinement.

Variant	Seed	AP	F1	Jaccard	DDI
Full model	0	0.7777	0.6466	0.4997	0.0595
	1	0.7768	0.6469	0.4999	0.0588
	2	0.7776	0.6510	0.5048	0.0602
	Mean ± *SD*	0.7774 ± 0.0005	0.6482 ± 0.0025	0.5015 ± 0.0029	0.0595 ± 0.0007
w/o longitudinal history	0	0.7323	0.6040	0.4530	0.0613
	1	0.7319	0.6039	0.4536	0.0602
	2	0.7324	0.6084	0.4580	0.0609
	Mean ± *SD*	0.7322 ± 0.0003	0.6054 ± 0.0026	0.4549 ± 0.0027	0.0608 ± 0.0006
w/o training-case memory	0	0.7667	0.6381	0.4898	0.0591
	1	0.7638	0.6389	0.4912	0.0592
	2	0.7664	0.6421	0.4941	0.0590
	Mean ± *SD*	0.7656 ± 0.0016	0.6397 ± 0.0021	0.4917 ± 0.0022	0.0591 ± 0.0001
w/o medication-set block	0	0.7746	0.6449	0.4984	0.0579
	1	0.7754	0.6460	0.4992	0.0578
	2	0.7756	0.6478	0.5009	0.0590
	Mean ± *SD*	0.7752 ± 0.0005	0.6462 ± 0.0015	0.4995 ± 0.0013	0.0582 ± 0.0006
w/o clinical associations	0	0.7538	0.6325	0.4843	0.0589
	1	0.7537	0.6324	0.4848	0.0579
	2	0.7559	0.6348	0.4873	0.0591
	Mean ± *SD*	0.7545 ± 0.0013	0.6332 ± 0.0014	0.4855 ± 0.0016	0.0586 ± 0.0006

Table S12: MIMIC-IV: component ablations after DDI-feedback optimization.

Variant	Seed	AP	F1	Jaccard	DDI
Full model	0	0.7767	0.6428	0.4961	0.0511
	1	0.7761	0.6455	0.4993	0.0518
	2	0.7760	0.6472	0.5011	0.0538
	Mean ± *SD*	**0.7763** ± 0.0004	**0.6452** ± 0.0022	**0.4988** ± 0.0026	**0.0522** ± 0.0014

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Table S12 (continued)

Variant	Seed	AP	F1	Jaccard	DDI
w/o longitudinal history	0	0.7313	0.6051	0.4551	0.0490
	1	0.7298	0.5993	0.4487	0.0467
	2	0.7306	0.6046	0.4553	0.0494
	Mean $\pm$ *SD*	0.7306 ± 0.0008	0.6030 ± 0.0032	0.4531 ± 0.0037	0.0484 ± 0.0014
w/o training-case memory	0	0.7664	0.6371	0.4899	0.0503
	1	0.7634	0.6354	0.4882	0.0527
	2	0.7671	0.6383	0.4911	0.0517
	Mean $\pm$ *SD*	0.7657 ± 0.0020	0.6369 ± 0.0015	0.4897 ± 0.0015	0.0516 ± 0.0012
w/o medication-set block	0	0.7742	0.6425	0.4959	0.0519
	1	0.7751	0.6442	0.4974	0.0520
	2	0.7748	0.6454	0.4988	0.0525
	Mean $\pm$ *SD*	0.7747 ± 0.0005	0.6440 ± 0.0015	0.4973 ± 0.0014	0.0521 ± 0.0004
w/o clinical associations	0	0.7523	0.6283	0.4812	0.0519
	1	0.7541	0.6271	0.4797	0.0511
	2	0.7558	0.6311	0.4836	0.0536
	Mean $\pm$ *SD*	0.7541 ± 0.0017	0.6288 ± 0.0020	0.4815 ± 0.0020	0.0522 ± 0.0012

Table S13: NWICU: component ablations after evidence refinement.

Variant	Seed	AP	F1	Jaccard	DDI
Full model	0	0.6904	0.6042	0.4530	0.1144
	1	0.6904	0.6053	0.4546	0.1163
	2	0.6907	0.6064	0.4558	0.1202
	Mean $\pm$ *SD*	0.6905 ± 0.0002	0.6053 ± 0.0011	0.4545 ± 0.0014	0.1170 ± 0.0029
w/o longitudinal history	0	0.5949	0.5108	0.3589	0.1333
	1	0.5948	0.5112	0.3590	0.1396
	2	0.5941	0.5076	0.3562	0.1315
	Mean $\pm$ *SD*	0.5946 ± 0.0004	0.5099 ± 0.0019	0.3580 ± 0.0016	0.1348 ± 0.0042
w/o training-case memory	0	0.6897	0.6005	0.4492	0.1150
	1	0.6889	0.6005	0.4489	0.1141
	2	0.6879	0.6011	0.4500	0.1173
	Mean $\pm$ *SD*	0.6888 ± 0.0009	0.6007 ± 0.0003	0.4494 ± 0.0006	0.1155 ± 0.0016
w/o medication-set block	0	0.6907	0.6031	0.4519	0.1129
	1	0.6914	0.6045	0.4535	0.1156
	2	0.6896	0.6035	0.4531	0.1177
	Mean $\pm$ *SD*	0.6906 ± 0.0009	0.6037 ± 0.0007	0.4528 ± 0.0008	0.1154 ± 0.0024
w/o clinical associations	0	0.6735	0.5921	0.4407	0.1176
	1	0.6761	0.5922	0.4412	0.1169
	2	0.6772	0.5928	0.4420	0.1197
	Mean $\pm$ *SD*	0.6756 ± 0.0019	0.5924 ± 0.0004	0.4413 ± 0.0007	0.1180 ± 0.0015

Table S14: NWICU: component ablations after DDI-feedback optimization.

Variant	Seed	AP	F1	Jaccard	DDI
Full model	0	0.6913	0.6036	0.4524	0.1077
	1	0.6901	0.6007	0.4497	0.1016
	2	0.6912	0.6034	0.4525	0.1035
	Mean ± *SD*	**0.6909** ± 0.0007	**0.6026** ± 0.0016	**0.4515** ± 0.0016	**0.1043** ± 0.0032
w/o longitudinal history	0	0.5944	0.5076	0.3560	0.1196
	1	0.5939	0.5084	0.3563	0.1124
	2	0.5933	0.5078	0.3563	0.1201
	Mean ± *SD*	0.5939 ± 0.0006	0.5079 ± 0.0004	0.3562 ± 0.0002	0.1173 ± 0.0043
w/o training-case memory	0	0.6896	0.5986	0.4473	0.1056
	1	0.6874	0.5941	0.4428	0.0893
	2	0.6876	0.5966	0.4452	0.0941
	Mean ± *SD*	0.6882 ± 0.0012	0.5964 ± 0.0022	0.4451 ± 0.0022	0.0963 ± 0.0084
w/o medication-set block	0	0.6915	0.6006	0.4495	0.1016
	1	0.6900	0.6007	0.4493	0.1000
	2	0.6904	0.6014	0.4505	0.1038
	Mean ± *SD*	0.6906 ± 0.0008	0.6009 ± 0.0004	0.4497 ± 0.0007	0.1018 ± 0.0019
w/o clinical associations	0	0.6733	0.5901	0.4390	0.1023
	1	0.6758	0.5869	0.4357	0.0948
	2	0.6777	0.5859	0.4346	0.0888
	Mean ± *SD*	0.6756 ± 0.0022	0.5876 ± 0.0022	0.4364 ± 0.0023	0.0953 ± 0.0067

Table S15: SICdb: component ablations after evidence refinement.

Variant	Seed	AP	F1	Jaccard	DDI
Full model	0	0.7506	0.6459	0.4986	0.1254
	1	0.7515	0.6480	0.5014	0.1280
	2	0.7502	0.6495	0.5028	0.1334
	Mean ± *SD*	0.7507 ± 0.0007	0.6478 ± 0.0018	0.5009 ± 0.0021	0.1289 ± 0.0041
w/o longitudinal history	0	0.7189	0.6099	0.4613	0.1384
	1	0.7194	0.6109	0.4632	0.1442
	2	0.7194	0.6110	0.4635	0.1416
	Mean ± *SD*	0.7192 ± 0.0003	0.6106 ± 0.0006	0.4626 ± 0.0012	0.1414 ± 0.0029
w/o training-case memory	0	0.7513	0.6474	0.5014	0.1349
	1	0.7475	0.6436	0.4973	0.1326
	2	0.7524	0.6486	0.5029	0.1380
	Mean ± *SD*	0.7504 ± 0.0026	0.6465 ± 0.0026	0.5005 ± 0.0029	0.1352 ± 0.0027
w/o medication-set block	0	0.7497	0.6475	0.5005	0.1259
	1	0.7521	0.6511	0.5044	0.1261
	2	0.7480	0.6469	0.5000	0.1232
	Mean ± *SD*	0.7499 ± 0.0021	0.6485 ± 0.0022	0.5017 ± 0.0024	0.1250 ± 0.0016

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Table S15 (continued)

Variant	Seed	AP	F1	Jaccard	DDI
w/o clinical associations	0	0.7449	0.6456	0.4986	0.1316
	1	0.7444	0.6465	0.4999	0.1321
	2	0.7438	0.6447	0.4974	0.1293
	Mean ± *SD*	0.7444 ± 0.0006	0.6456 ± 0.0009	0.4986 ± 0.0012	0.1310 ± 0.0015

Table S16: SICdb: component ablations after DDI-feedback optimization.

Variant	Seed	AP	F1	Jaccard	DDI
Full model	0	0.7497	0.6414	0.4943	0.1125
	1	0.7517	0.6408	0.4928	0.1194
	2	0.7502	0.6451	0.4979	0.1225
	Mean ± *SD*	**0.7505** ± 0.0010	**0.6425** ± 0.0023	**0.4950** ± 0.0026	**0.1181** ± 0.0052
w/o longitudinal history	0	0.7203	0.6082	0.4599	0.1131
	1	0.7193	0.6091	0.4614	0.1208
	2	0.7196	0.6112	0.4637	0.1210
	Mean ± *SD*	0.7197 ± 0.0005	0.6095 ± 0.0015	0.4617 ± 0.0019	0.1183 ± 0.0045
w/o training-case memory	0	0.7522	0.6452	0.4989	0.1215
	1	0.7484	0.6439	0.4970	0.1230
	2	0.7516	0.6421	0.4957	0.1199
	Mean ± *SD*	0.7507 ± 0.0021	0.6437 ± 0.0015	0.4972 ± 0.0016	0.1215 ± 0.0015
w/o medication-set block	0	0.7506	0.6455	0.4984	0.1168
	1	0.7491	0.6424	0.4950	0.1170
	2	0.7475	0.6412	0.4939	0.1130
	Mean ± *SD*	0.7491 ± 0.0016	0.6430 ± 0.0022	0.4958 ± 0.0024	0.1156 ± 0.0023
w/o clinical associations	0	0.7444	0.6443	0.4972	0.1266
	1	0.7441	0.6428	0.4950	0.1208
	2	0.7445	0.6440	0.4964	0.1264
	Mean ± *SD*	0.7443 ± 0.0002	0.6437 ± 0.0008	0.4962 ± 0.0011	0.1246 ± 0.0033

S4.4 Fixed and dynamic DDI regularization

The fixed-weight controls and dynamic rule share the initial refinement model, optimizer settings, 20-epoch budget and validation criteria. Fixed weight 1 selects the initial refinement model in all nine runs under these criteria. Repeated values represent the selected models.

Table S17: MIMIC-IV: fixed versus dynamic DDI regularization.

Variant	Seed	AP	F1	Jaccard	DDI
Fixed weight 0	0	0.7775	0.6453	0.4987	0.0574
	1	0.7771	0.6440	0.4973	0.0570
	2	0.7761	0.6482	0.5026	0.0578
	Mean ± *SD*	0.7769 ± 0.0007	0.6459 ± 0.0021	0.4995 ± 0.0027	0.0574 ± 0.0004

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Table S17 (continued)

Variant	Seed	AP	F1	Jaccard	DDI
Fixed weight 0.05	0	0.7763	0.6446	0.4986	0.0522
	1	0.7761	0.6423	0.4957	0.0520
	2	0.7766	0.6479	0.5012	0.0541
	Mean ± *SD*	0.7763 ± 0.0002	0.6449 ± 0.0029	0.4985 ± 0.0028	0.0528 ± 0.0012
Fixed weight 0.1	0	0.7762	0.6425	0.4959	0.0491
	1	0.7753	0.6431	0.4960	0.0486
	2	0.7763	0.6473	0.5008	0.0496
	Mean ± *SD*	0.7759 ± 0.0005	0.6443 ± 0.0026	0.4975 ± 0.0028	0.0491 ± 0.0005
Fixed weight 1	0	0.7777	0.6466	0.4997	0.0595
	1	0.7768	0.6469	0.4999	0.0588
	2	0.7776	0.6510	0.5048	0.0602
	Mean ± *SD*	0.7774 ± 0.0005	0.6482 ± 0.0025	0.5015 ± 0.0029	0.0595 ± 0.0007
Dynamic weight (EviDrug)	0	0.7767	0.6428	0.4961	0.0511
	1	0.7761	0.6455	0.4993	0.0518
	2	0.7760	0.6472	0.5011	0.0538
	Mean ± *SD*	**0.7763** ± 0.0004	**0.6452** ± 0.0022	**0.4988** ± 0.0026	**0.0522** ± 0.0014

Table S18: NWICU: fixed versus dynamic DDI regularization.

Variant	Seed	AP	F1	Jaccard	DDI
Fixed weight 0	0	0.6914	0.6034	0.4518	0.1118
	1	0.6913	0.6014	0.4499	0.1119
	2	0.6921	0.6036	0.4523	0.1144
	Mean ± *SD*	0.6916 ± 0.0004	0.6028 ± 0.0012	0.4513 ± 0.0013	0.1127 ± 0.0015
Fixed weight 0.05	0	0.6908	0.6033	0.4521	0.1032
	1	0.6911	0.5997	0.4484	0.0982
	2	0.6925	0.6021	0.4509	0.1015
	Mean ± *SD*	0.6915 ± 0.0009	0.6017 ± 0.0018	0.4505 ± 0.0019	0.1010 ± 0.0025
Fixed weight 0.1	0	0.6900	0.6019	0.4508	0.1013
	1	0.6904	0.6002	0.4488	0.1034
	2	0.6900	0.6034	0.4525	0.1037
	Mean ± *SD*	0.6901 ± 0.0003	0.6018 ± 0.0016	0.4507 ± 0.0018	0.1028 ± 0.0013
Fixed weight 1	0	0.6904	0.6042	0.4530	0.1144
	1	0.6904	0.6053	0.4546	0.1163
	2	0.6907	0.6064	0.4558	0.1202
	Mean ± *SD*	0.6905 ± 0.0002	0.6053 ± 0.0011	0.4545 ± 0.0014	0.1170 ± 0.0029
Dynamic weight (EviDrug)	0	0.6913	0.6036	0.4524	0.1077
	1	0.6901	0.6007	0.4497	0.1016
	2	0.6912	0.6034	0.4525	0.1035
	Mean ± *SD*	**0.6909** ± 0.0007	**0.6026** ± 0.0016	**0.4515** ± 0.0016	**0.1043** ± 0.0032

Table S19: SICdb: fixed versus dynamic DDI regularization.

Variant	Seed	AP	F1	Jaccard	DDI
Fixed weight 0	0	0.7510	0.6446	0.4975	0.1252
	1	0.7515	0.6436	0.4959	0.1251
	2	0.7514	0.6472	0.5001	0.1278
	Mean ± *SD*	0.7513 ± 0.0002	0.6451 ± 0.0019	0.4978 ± 0.0021	0.1260 ± 0.0015
Fixed weight 0.05	0	0.7503	0.6411	0.4939	0.1187
	1	0.7509	0.6413	0.4935	0.1191
	2	0.7521	0.6455	0.4984	0.1229
	Mean ± *SD*	0.7511 ± 0.0009	0.6426 ± 0.0025	0.4953 ± 0.0027	0.1202 ± 0.0023
Fixed weight 0.1	0	0.7503	0.6407	0.4932	0.1152
	1	0.7506	0.6416	0.4941	0.1185
	2	0.7508	0.6457	0.4988	0.1199
	Mean ± *SD*	0.7506 ± 0.0003	0.6427 ± 0.0027	0.4954 ± 0.0030	0.1179 ± 0.0024
Fixed weight 1	0	0.7506	0.6459	0.4986	0.1254
	1	0.7515	0.6480	0.5014	0.1280
	2	0.7502	0.6495	0.5028	0.1334
	Mean ± *SD*	0.7507 ± 0.0007	0.6478 ± 0.0018	0.5009 ± 0.0021	0.1289 ± 0.0041
Dynamic weight (EviDrug)	0	0.7497	0.6414	0.4943	0.1125
	1	0.7517	0.6408	0.4928	0.1194
	2	0.7502	0.6451	0.4979	0.1225
	Mean ± *SD*	**0.7505** ± 0.0010	**0.6425** ± 0.0023	**0.4950** ± 0.0026	**0.1181** ± 0.0052

S5 Complete SICdb Medication Mapping

The dictionary contains 520 medication terms: 303 have an accepted ATC3 class and 217 remain unresolved. The rule labels below distinguish accepted mappings (M1–M4) from unresolved terms (M5–M7). These are dictionary entries, without patient counts or clinical records.

Table S20: SICdb dictionary mapping rules.

Rule	Decision	Terms
M1	Exact generic ingredient	150
M2	Explicit salt or terminal-e orthographic alias	140
M3	Multiple ingredients with one common ATC3 class	10
M4	Unique RxNorm ingredient with direct ATC membership	3
M5	No uniquely identified ingredient	103
M6	Ingredient maps to multiple ATC3 classes	98
M7	Combination without a unique class	16

Table S21: All SICdb medication-dictionary decisions.

Medication dictionary term	ATC3	Rule
Abciximab	B01A	M1
Acenocoumarol	B01A	M1
Acetazolamid	S01E	M2
Acetylcystein	Unresolved	M6
Acetylsalicylat	Unresolved	M5
Acetylsalicyls	Unresolved	M5
Acyclovir	Unresolved	M5
ADENOsin	C01E	M2
Agomelatin	N06A	M2
Aktivkohle	Unresolved	M5
Alfentanil	N01A	M1
Allopurinol	M04A	M1
Alprazolam	N05B	M1
Alprostadil	Unresolved	M6
Alteplase	Unresolved	M6
Amantadinsulfat	Unresolved	M5
Amikacin	Unresolved	M6
Amilorid+HCT	Unresolved	M7
AMIOdaron	C01B	M2
Amitriptylin	N06A	M2
Amlodipin	C08C	M2
Amlodipin besilat+Valsartan+HCT	Unresolved	M6
Amoxicillin/Clavulans	Unresolved	M7
Amphotericin B	Unresolved	M6
Ampicillin	Unresolved	M6
Ampicillin/Sulbactam	J01C	M3
Anastrozol	L02B	M2
Anidulafungin	J02A	M1
Antithrombin III	B01A	M1
Apixaban	B01A	M1
Argatroban	B01A	M1
Aripiprazol	N05A	M2
Atenolol	C07A	M1
Atorvastatin	C10A	M1
Atropin sulfuricum	Unresolved	M6
Azathioprin	L04A	M2
Azithromycin	Unresolved	M6
Baclofen	M03B	M1
Beclometason und Formeterol	Unresolved	M6
Benzydamin	Unresolved	M6
Bezafibrat	C10A	M2
Bicalutamid	L02B	M2
Biperidin	Unresolved	M5
Biperidin Ret.	Unresolved	M5
Bisacodyl	A06A	M1
Bisoprolol	C07A	M1
Bisoprolol fumarat	C07A	M1
Bisoprolol+HCT	Unresolved	M7

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Table S21 (continued)

Medication dictionary term	ATC3	Rule
Bisoprolol-hemi-fumarat	C07A	M1
Bosentan-1-Wasser	C02K	M1
Brinzolamid	S01E	M2
Bromazepam	N05B	M1
Brotizolam	N05C	M1
Budesonid	Unresolved	M6
Budesonid und Formeterol	Unresolved	M6
BUPivacain 0.5%	N01B	M2
BUPivacain 0.5% hyperbar	N01B	M2
Buprenorphin	Unresolved	M6
Bupropion	N06A	M1
Cabergolin	Unresolved	M6
Cafedrin-Theodrenalin	C01C	M3
Calcitriol	Unresolved	M6
Calcium gluc.	Unresolved	M5
Calcium/Vit D	Unresolved	M5
Calciumglukonat 10%	Unresolved	M5
Candesartan	C09C	M1
Candesartan +HCT	Unresolved	M7
Captopril	C09A	M1
Carbamazepin	N03A	M2
Carbetocin	H01B	M1
Caroverin	A03A	M2
Carvedilol	C07A	M1
Caspofungin	J02A	M1
Cefazolin	J01D	M1
Cefepim	J01D	M2
Cefotaxim	J01D	M2
Ceftazidim	J01D	M2
Ceftriaxon	J01D	M2
Cefuroxim	Unresolved	M6
Celecoxib	Unresolved	M6
Cetrimid und Carbopol	Unresolved	M6
Cetylpyridinium	Unresolved	M6
Cetylpyridinium chlorid 1-Wasser	Unresolved	M6
Chlorhexidin	Unresolved	M6
Chloroquin	P01B	M2
Chlorprothixen	N05A	M2
Ciclesonid	Unresolved	M6
Ciclosporin	Unresolved	M6
Cilazapril	C09A	M1
Cinacalcet hydrochlorid	H05B	M1
Ciprofloxacin	Unresolved	M6
Cisatracurium	M03A	M1
Citalopram	N06A	M1
Clarithromycin	J01F	M1
CLEMAstin	Unresolved	M6
Clindamycin	Unresolved	M6
Clobazam	N05B	M1

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Table S21 (continued)

Medication dictionary term	ATC3	Rule
Clomipramin	N06A	M2
ClonazePAM	N03A	M1
CloNIDin	Unresolved	M6
Clopidogrel	B01A	M1
Colistimethat	Unresolved	M5
Colistin	Unresolved	M6
Cyanocobolamin	Unresolved	M5
Cyclophosphamid	L01A	M2
Cyclosporin A	Unresolved	M5
Dabigatran	Unresolved	M5
Dalteparin	B01A	M1
Danaparoid	B01A	M1
Dapagliflozin	A10B	M1
Dapagliflozin+Metformin hydrochlorid 5/1000	A10B	M3
Dapagliflozin+Metformin hydrochlorid 5/850	A10B	M3
Daptomycin	J01X	M1
Desloratadin	R06A	M2
DesmoPRESSINazetat	Unresolved	M5
DEXAmethason	Unresolved	M6
Dexmedetomidin	N05C	M2
Dexpanthenol	Unresolved	M6
Diazepam	N05B	M1
Diclofenac	Unresolved	M6
Diclofenac+Orphenadrincitrat	Unresolved	M6
Digitoxin	C01A	M1
DihydrALAZIN	C02D	M2
Dihydrocodein	N02A	M2
Diltiazem	Unresolved	M6
Dimetinden maleat	Unresolved	M6
Diphenhydramin	Unresolved	M6
Distigmin	N07A	M2
DOBUtamin	C01C	M2
Domperidon	A03F	M2
Donepezil	N06D	M1
DOPamin	C01C	M2
Doxazosin	C02C	M1
Doxycyclin	Unresolved	M6
Doxycyclin monohydrat	Unresolved	M6
Dronedaron hydrochlorid	C01B	M2
Droperidol	N05A	M1
Duloxetin	N06A	M2
Dutasterid	G04C	M2
Econazol	Unresolved	M6
Efavirenz	J05A	M1
Eisen II	Unresolved	M5
Eisen Saccharose	Unresolved	M5
Enalapril	C09A	M1
Enalapril maleat	C09A	M1
Enalapril maleat+HCT	Unresolved	M7

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Table S21 (continued)

Medication dictionary term	ATC3	Rule
Enalapril+HCT	Unresolved	M7
Enoxaparin prophylakt.	B01A	M1
Enoxaparin therapeut.	B01A	M1
Ephedrin	Unresolved	M6
EPINEPHrin	Unresolved	M6
Eplerenon	C03D	M2
Eprosartan	C09C	M1
Ertapenem	J01D	M1
Erythromycin	Unresolved	M6
Erythropoetin	Unresolved	M5
Escitalopram	N06A	M1
Esmolol	C07A	M1
Esomeprazol	A02B	M2
Etacrynsaeure	Unresolved	M5
Etilefrin	C01C	M2
Etomidate	N01A	M1
Everolimus	Unresolved	M6
Excicial Hydrocreme	Unresolved	M5
Exemestan	L02B	M2
Extractum fructus silybi mariani siccum	Unresolved	M5
Ezetimib	C10A	M2
Ezetimib+Simvastatin	C10A	M3
Faktor VIIa	Unresolved	M5
Fenofibrat	C10A	M2
FentaNYL	Unresolved	M6
Fibrinogen	Unresolved	M5
Filgrastim	L03A	M1
Finasterid	Unresolved	M6
Fisch	Unresolved	M5
Flavonoidfraktion	Unresolved	M5
Flucloxacillin	J01C	M1
Fluconazol	Unresolved	M6
Fludrocortison	H02A	M2
Flumazenil	V03A	M1
Flunitrazepam	N05C	M1
Fluoxetin	N06A	M2
Flupentixol+Melitracen	Unresolved	M6
Fluticason	Unresolved	M6
Fluvastatin	C10A	M1
Fols	Unresolved	M5
Fondaparinux	B01A	M1
Formoterol-fumarat-dihydrat	R03A	M1
Fosfomycin	J01X	M1
Fosinopril Natrium	C09A	M1
FUROsemid	C03C	M2
Furosemid+Spironolacton	Unresolved	M6
Fusidins	Unresolved	M5
Gabapentin	N03A	M1
Galantamin	N06D	M2

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Table S21 (continued)

Medication dictionary term	ATC3	Rule
Ganciclovir	Unresolved	M6
Gentamycin	Unresolved	M5
Gentamycinsulfat	Unresolved	M5
Gerinnungsfaktor XIII	Unresolved	M5
Gerinnungsfaktoren II, VI, IX, X	Unresolved	M5
Ginkgoblatt Trockenextrakt	Unresolved	M5
Gliclazid	A10B	M2
Glimepirid	A10B	M2
Gliquidon	A10B	M2
Glucagon hydrochlorid	H04A	M1
Glucose-1-Phosphat	Unresolved	M6
Glutamin	A16A	M2
Glyceroltrinitrat	Unresolved	M5
Glycopyrronium	Unresolved	M6
Glyzeroltrinitrat	Unresolved	M5
Haloperidol	N05A	M1
Heparin	Unresolved	M6
HuG-CSF	Unresolved	M5
Hydrochlorothiazid	C03A	M2
Hydrochlorothiazid+Ramipril	Unresolved	M6
Hydrocortison	Unresolved	M6
HYDROmorphon	N02A	M2
HYDROmorphon retard	N02A	M2
Hydroxycarbamid	L01X	M2
Hydroxyzin	N05B	M2
Hyoscin-N-Butylbromid	Unresolved	M5
Ibuprofen	Unresolved	M6
Idarucizumab	V03A	M1
Iloprost	B01A	M1
Imatinib mesilat	L01E	M1
Imipenem/Cilastin	Unresolved	M5
Immunglobulin	Unresolved	M5
Indapamid	C03B	M2
Insulin	A10A	M4
Insulin glargin	A10A	M4
Insulin lispro	A10A	M1
Insulin-Aspart-Protamin-Kristall 30/70	Unresolved	M6
Insulin-Aspart-Protamin-Kristall 50/50	Unresolved	M6
Insulin-Aspart-Protamin-Kristall 70/30	Unresolved	M6
Insulin-Isophan	Unresolved	M5
Ipratroprium und Fenoterol	Unresolved	M6
Ipratropriumbromid	Unresolved	M5
Isoniazid	J04A	M1
Isoprenalin	Unresolved	M6
Ivabradin	C01E	M2
Kalium	Unresolved	M5
Kalium jodid+Levothyroxin natrium	Unresolved	M7
Kaliumcanrenoat	Unresolved	M5
Kaliumchlorid	Unresolved	M5

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Table S21 (continued)

Medication dictionary term	ATC3	Rule
KETAmin	N01A	M2
Kybernin P	Unresolved	M5
Labetalol	C07A	M1
Lacosamid	N03A	M2
Lactobacillus casei	G01A	M1
Lamotrigin	N03A	M2
Latanoprost	S01E	M1
Leflunomid	L04A	M2
Lercanidipin	C08C	M2
LevETIRAcetam	N03A	M1
Levocetirizin	R06A	M2
Levodopa +Benserazid	Unresolved	M7
Levodopa+Benserazid	Unresolved	M7
Levodopa/Carbidopa/Entacapon	N04B	M3
Levofloxacin	Unresolved	M6
Levomepromazin	N05A	M2
Levosimendan	C01C	M1
LEVOthyroxin	H03A	M2
Lidocain 1%	Unresolved	M6
Lidocain 2%	Unresolved	M6
Linagliptin	A10B	M1
Linezolid	J01X	M1
Liposomales Amphotericin B	Unresolved	M6
Liraglutid	A10B	M2
Lisinopril	C09A	M1
Lisinopril HCT	C09A	M1
Lithium carbonat	N05A	M1
LORazepam	N05B	M1
Losartan	C09C	M1
Losartan+HCT	Unresolved	M7
Macrogol	A06A	M1
Macrogol 3350	A06A	M1
Magnesium	Unresolved	M5
Magnesiumgluconicum	Unresolved	M5
Mefenamins	Unresolved	M5
Melatonin	N05C	M1
Meloxicam	M01A	M1
Melperon hydrochlorid	N05A	M2
Memantin	N06D	M2
Mepivacain 1%	N01B	M2
Mepivacain 1,5%	N01B	M2
Mepivacain 2%	N01B	M2
Meropenem	J01D	M1
Mesalazin	A07E	M2
Mesna	Unresolved	M6
Metamizol	N02B	M2
Metamizol gtt	N02B	M2
Metformin	A10B	M1
Metformin +Vildagliptin	A10B	M3

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Table S21 (continued)

Medication dictionary term	ATC3	Rule
Metformin hydrochlorid	A10B	M1
Metformin+Sitagliptin	A10B	M3
Methadon	N07B	M2
Methohexital	Unresolved	M6
Methotrexat	Unresolved	M6
Methylenblau	Unresolved	M5
Methylnaltrexon	A06A	M2
Methylphenidat hydrochlorid	N06B	M2
MethylPREDNIsolon	Unresolved	M6
Metildigoxin	C01A	M1
Metoclopramid	A03F	M2
Metoprolol	C07A	M1
Metronidazol	Unresolved	M6
Micafungin	J02A	M1
Miconazol	Unresolved	M6
Midazolam	N05C	M1
Milnacipran hydrochlorid	N06A	M1
Minoxidil	Unresolved	M6
Mirtazepin	Unresolved	M5
Mivacurim	Unresolved	M5
Moclobemid	N06A	M2
Morphin	N02A	M2
Morphinsulfat	Unresolved	M5
Moxifloxacin	Unresolved	M6
Moxonidin	C02A	M2
Mycophenolat	Unresolved	M5
Mycophenolat mofetil	Unresolved	M5
Mycostatin Zinkoxid Paste	Unresolved	M5
N-Acetylcystein	Unresolved	M6
Nalbuphin	N02A	M2
Naloxon	Unresolved	M6
Naproxen Saft	Unresolved	M6
Naproxen TBL	Unresolved	M6
Natrium hydrogencarbonat	Unresolved	M5
Natrium perchlorat	Unresolved	M5
Natriumchlorid Sachet	Unresolved	M5
Natriumglyzerophosphat	Unresolved	M5
Natriumhydrogencarbonat	Unresolved	M5
Natriumnitroprussid	Unresolved	M5
Natriumpicosulfat	Unresolved	M5
Natriumpolystyrensulfonat	Unresolved	M5
Natriumvalproat	Unresolved	M5
Nebivolol	C07A	M1
Neostigmin	Unresolved	M6
Nicht gelistete Inhalation	Unresolved	M5
Nicht gelistete Kurzinfusion	Unresolved	M5
Nicht gelistete Salbe/Creme	Unresolved	M5
Nicht gelistete Tropfen	Unresolved	M5
Nicht gelisteter Perfusor	Unresolved	M5

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Table S21 (continued)

Medication dictionary term	ATC3	Rule
Nicht gelistetes anderes Medikament	Unresolved	M5
Nicht gelistetes Antiinfektivum	Unresolved	M5
Nicht gelistetes Bolusmedikament	Unresolved	M5
Nicht gelistetes Gerinnungsmedikament	Unresolved	M5
Nicht gelistetes Gerinnungsmedikament IU	Unresolved	M5
Nicht gelistetes orales Medikament	Unresolved	M5
Nicht gelistetes sc Medikament IU	Unresolved	M5
Nicht gelistetes sc Medikament mg	Unresolved	M5
Nicorandil	C01D	M1
Nicotin	N07B	M2
Nicotinamid+Thiamin+Riboflavin	Unresolved	M6
NIFEdipin	C08C	M2
NiMODipin	C08C	M2
Nitrendipin	C08C	M2
Nitrendipin +Enalapril	Unresolved	M6
Nitroglycerin	Unresolved	M5
Norepinephrin	C01C	M2
ocTREOtid	H01C	M2
Oculotect Fluid Au-Tr	Unresolved	M5
Ofloxacin	Unresolved	M6
Olanzapin	N05A	M2
Olmesartan	Unresolved	M5
Olmesartan+Amlodipin	Unresolved	M7
Omnibiotic AAD	Unresolved	M5
On	Unresolved	M5
ONDANsetron	A04A	M1
Ornithinaspartat	Unresolved	M5
Oseltamivir	J05A	M1
Oxazepam	N05B	M1
Oxcarbazepin	N03A	M2
Oxycodon	N02A	M2
Oxycodon ret	N02A	M2
Oxytocin	H01B	M1
Pankreatin	Unresolved	M5
Pantoprazol	A02B	M2
Paracetamol	N02B	M1
Paromomycin	A07A	M1
Paroxetinhydrochlorid-Hemihydrat	Unresolved	M5
Passionsblumenkraut	Unresolved	M5
Penicillin G	Unresolved	M5
Pentoxifyllin	C04A	M2
Perampanel	N03A	M1
Pethidin	N02A	M2
Phenprocoumon	B01A	M1
Phenylephrin	Unresolved	M6
Phenytoin	N03A	M1
PHYSostigmin	Unresolved	M6
Phytmenadion	Unresolved	M5
Piperacillin/Tazobactam	J01C	M3

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Table S21 (continued)

Medication dictionary term	ATC3	Rule
Piracetam	N06B	M1
PIRItramid	N02A	M2
Posaconazol	J02A	M2
Pramipexol	N04B	M2
Prasugrel	B01A	M1
Pravastatin	C10A	M1
PredniSOLON	Unresolved	M6
Pregabalin	N03A	M1
Prilocain 1%	N01B	M2
Primidon	N03A	M2
Promethazin	Unresolved	M6
Propafenon	C01B	M2
Propofol 1%	N01A	M1
Propofol 2%	N01A	M1
Propranolol	C07A	M1
Protaminsulfat	Unresolved	M5
Prothipendyl	N05A	M1
Protifar	Unresolved	M5
Pyrazinamid	J04A	M2
Pyridostigmin bromid	N07A	M2
Quetiapin	N05A	M2
Ramipril	C09A	M1
Ramipril/HCT	Unresolved	M7
Ranitidin	A02B	M2
Rasagilin	N04B	M2
Remifentanil	N01A	M1
Retinol	Unresolved	M6
Retinol und Dexpantenol	Unresolved	M6
Rifampicin	J04A	M1
Rifaximin	Unresolved	M6
Rilmenidin	C02A	M2
Rilmenidin dihydrogenphosphat	C02A	M2
Risperidon	N05A	M2
Rivaroxaban	B01A	M1
Rivastigmin	N06D	M2
ROCuronium	M03A	M2
Roflumilast	R03D	M1
Ropinirol	N04B	M2
ROPIvacain	N01B	M2
ROPIvacain 0.15%	N01B	M2
ROPIvacain 0.2%	N01B	M2
ROPIvacain 0.375%	N01B	M2
ROPIvacain 0.5%	N01B	M2
ROPIvacain 0.75%	N01B	M2
ROPIvacain 1%	N01B	M2
Rosuvastatin	C10A	M1
Rotigotin	N04B	M2
S-Ketamin	N01A	M2
Salbutamol	Unresolved	M6

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Table S21 (continued)

Medication dictionary term	ATC3	Rule
Salmeterol	R03A	M1
Salmeterol/Fluticason	Unresolved	M6
Scopolamin	Unresolved	M6
Sertralin	N06A	M2
Sevelamer	V03A	M1
Sevelamercarbonat	Unresolved	M5
Sevofluran (Anaconda)	N01A	M2
Sildenafil	G04B	M1
Simeticon	Unresolved	M5
Simvastatin	C10A	M1
Sitagliptin phosphat 1 H2O	A10B	M1
Somatostatin	H01C	M1
Sotalol	C07A	M1
Spironolacton	C03D	M2
Succinylcholin	Unresolved	M5
Sucralfat	A02B	M2
SUFentanil	N01A	M1
Sugammadex	V03A	M1
Sulfametoxazol+Trimethoprim	Unresolved	M7
Tacrolimus	Unresolved	M6
Tamsulosin	G04C	M1
Tamsulosin retard	G04C	M1
Tantum verde Lutschtabletten	Unresolved	M5
Teicoplanin	J01X	M1
Telmisartan	C09C	M1
Tenecteplase	B01A	M1
Tenofovir disoproxil+Emtricitabin	J05A	M3
Terbinafin	Unresolved	M6
Terbutalin	Unresolved	M6
Terlipressin	H01B	M1
Tetanus Toxoid	J07A	M1
Tetracosactid	H01A	M2
Tetrazepam	M03B	M1
Theophyllin	R03D	M2
Thiamazol	H03B	M2
Thiamin	A11D	M4
Thiamin+Pyridoxin+Cyanocobalamin	Unresolved	M7
Thiaminhydrochlorid	Unresolved	M5
Thiopentalnatrium	Unresolved	M5
Tiaprid	N05A	M2
Ticagrelor	B01A	M1
Tigecyclin	J01A	M2
Tiotropium	R03B	M2
Tiotropium Lsg 2,5	R03B	M2
Tiotropium+Olodaterol 2,5/2,5	Unresolved	M6
Tizanidin	M03B	M2
Tolterodin-L-tartrat	G04B	M2
Topiramat	N03A	M2
Torasemid	C03C	M2

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Table S21 (continued)

Medication dictionary term	ATC3	Rule
traMADol	N02A	M1
Tranexams	Unresolved	M5
Trazodon ret	N06A	M2
Triazolam	N05C	M1
Trimethoprim/Sulfametoxazol	Unresolved	M7
Trometamol	Unresolved	M6
Trospium chlorid	G04B	M1
Urapidil	C02C	M1
Urokinase	B01A	M1
Ursodeoxychols	Unresolved	M5
Valaciclovir	J05A	M1
Valganciclovir	J05A	M1
Valproins	Unresolved	M5
Valsartan	C09C	M1
Valsartan+Amlodipin	Unresolved	M6
Valsartan+HCT	Unresolved	M7
Vancomycin	Unresolved	M6
Vasopressin	Unresolved	M5
Venlafaxin	N06A	M2
Verapamil	C08D	M1
Vildagliptin	A10B	M1
Vitamin D3	Unresolved	M5
Voriconazol	J02A	M2
Xipamid	C03B	M2
Ziprasidon	N05A	M2
Zolpidem	N05C	M1
Zolpidem tartrat 2	N05C	M1
Zuclopenthixol decanoat	N05A	M1
ZzColistin	Unresolved	M5
ZzRamipril HCT	Unresolved	M5
ZzRamipril/HCT	Unresolved	M5

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